

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                         |
|----------------------|-------------------------|
| United States Patent | 12399238                |
| Kind Code            | B2                      |
| Date of Patent       | August 26, 2025         |
| Inventor(s)          | Kojima; Hidekazu et al. |

---

### Magnetic sensor including magnetic detection element disposed on inclined surface

---

#### Abstract

A magnetic sensor includes a substrate including a top surface, an insulating layer including an inclined surface, an MR element disposed on the inclined surface, a first insulating portion of an insulating material disposed on a part of the MR element, and a second insulating portion of an insulating material disposed on another part of the MR element at a position forward of the first insulating portion in a direction along the inclined surface, the direction being a direction away from the top surface of the substrate.

---

|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
| <b>Inventors:</b> | <b>Kojima; Hidekazu (Tokyo, JP), Umehara; Hiromichi (Tokyo, JP)</b> |
| <b>Applicant:</b> | <b>TDK CORPORATION (Tokyo, JP)</b>                                  |
| <b>Family ID:</b> | <b>1000008782881</b>                                                |
| <b>Assignee:</b>  | <b>TDK CORPORATION (Tokyo, JP)</b>                                  |
| <b>Appl. No.:</b> | <b>18/751610</b>                                                    |
| <b>Filed:</b>     | <b>June 24, 2024</b>                                                |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240345188 A1          | Oct. 17, 2024           |

#### Foreign Application Priority Data

|    |             |               |
|----|-------------|---------------|
| JP | 2022-139173 | Sep. 01, 2022 |
|----|-------------|---------------|

#### Related U.S. Application Data

Publication Classification

Int. Cl.: G01R33/09 (20060101)

U.S. Cl.:

CPC G01R33/093 (20130101);

Field of Classification Search

CPC: G01R (33/0029); G01R (33/0052); G01R (33/0094); G01R (33/0206); G01R (33/093); G01R (33/098); G01R (33/09); G01R (33/072); G01R (33/091); G01R (33/02); G01R (15/207); G01R (33/0005); G01R (33/0047); G01R (33/063); G01R (1/06733); G01R (33/4808); G01R (33/445); H01L (24/33); H01L (24/49); H01L (2224/73215); H01L (22/12); H01L (25/0657); H01L (24/83); H01L (2224/16145); H01L (21/56); H01L (24/11); H01L (2224/73253); H01L (22/14); H01L (2224/83)

References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name   | U.S. Cl. | CPC          |
|--------------|-------------|-----------------|----------|--------------|
| 5227761      | 12/1992     | Sugimoto et al. | N/A      | N/A          |
| 7687284      | 12/2009     | Naito et al.    | N/A      | N/A          |
| 8178361      | 12/2011     | Naito et al.    | N/A      | N/A          |
| 11874346     | 12/2023     | Kojima          | N/A      | G01R 33/0017 |
| 11892527     | 12/2023     | Kojima          | N/A      | G01R 33/098  |
| 11897728     | 12/2023     | Kattainen       | N/A      | B66B 5/185   |
| 12000910     | 12/2023     | Umehara         | N/A      | G01R 33/0047 |
| 12044754     | 12/2023     | Kojima          | N/A      | G01R 33/0206 |
| 12061247     | 12/2023     | Watanabe        | N/A      | G01R 33/093  |
| 12188996     | 12/2024     | Ota             | N/A      | G01R 33/0206 |
| 2006/0176142 | 12/2005     | Naito et al.    | N/A      | N/A          |
| 2007/0080381 | 12/2006     | Chien et al.    | N/A      | N/A          |
| 2008/0032423 | 12/2007     | Wang et al.     | N/A      | N/A          |
| 2009/0268349 | 12/2008     | Makino et al.   | N/A      | N/A          |
| 2009/0273863 | 12/2008     | Kawano et al.   | N/A      | N/A          |
| 2012/0272514 | 12/2011     | Naito et al.    | N/A      | N/A          |
| 2021/0302511 | 12/2020     | Makino et al.   | N/A      | N/A          |

FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 2004-179250 | 12/2003          | JP      | N/A |
| 2004-319725 | 12/2003          | JP      | N/A |
| 2008-141210 | 12/2007          | JP      | N/A |

OTHER PUBLICATIONS

*Primary Examiner:* Nguyen; Vinh P

*Attorney, Agent or Firm:* Oliff PLC

---

## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/947,770, filed on Sep. 19, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/246,428 filed on Sep. 21, 2021, and Japanese Priority Patent Application No. 2022-139173 filed on Sep. 1, 2022, the entire contents of each of which are incorporated herein by their reference.

### **BACKGROUND**

(1) The technology relates to a magnetic sensor including magnetoresistive elements each disposed on an inclined surface.

(2) Magnetic sensors using magnetoresistive elements have been used for various applications in recent years. A system including a magnetic sensor may be intended to detect a magnetic field containing a component in a direction perpendicular to the surface of a substrate by using a magnetoresistive element provided on the substrate. In such a case, the magnetic field containing the component in the direction perpendicular to the surface of the substrate can be detected by providing a soft magnetic body for converting a magnetic field in the direction perpendicular to the surface of the substrate into a magnetic field in the direction parallel to the surface of the substrate or locating the magnetoresistive element on an inclined surface formed on the substrate.

(3) As the magnetoresistive elements, spin-valve magnetoresistive elements are used, for example. The spin-valve magnetoresistive element includes a magnetization pinned layer having a magnetization whose direction is fixed, a free layer having a magnetization whose direction is variable depending on the direction of an applied magnetic field, and a gap layer disposed between the magnetization pinned layer and the free layer.

(4) U.S. Patent Application Publication No. 2006/0176142 A1 discloses a magnetic sensor including magnetoresistive elements each formed on an inclined surface. Japanese Patent Application Laid-Open Publication No. 2008-141210 discloses a technique of forming two protective films of different materials on each side surface of a magnetoresistive element, thereby reducing a stress applied to the magnetoresistive element.

(5) Typically, in a case where a magnetoresistive element is formed on an inclined surface as in the magnetic sensor disclosed in U.S. Patent Application Publication No. 2006/0176142 A1, side surfaces of the magnetoresistive element are tapered. Herein, regarding a case where a spin-valve magnetoresistive element is used as a magnetoresistive element, suppose a case where the characteristics of the magnetoresistive element are controlled using an insulating layer formed around the magnetoresistive element as in the technique disclosed in Japanese Patent Application Laid-Open Publication No. 2008-141210. A first layer located close to the inclined surface and a second layer located away from the inclined surface have different areas. Therefore, influence of the insulating layer on the first layer and influence of the insulating layer on the second layer are also different. Consequently, the magnetoresistive element may undesirably have characteristics different from the intended ones.

### **SUMMARY**

(6) A magnetic sensor according to one embodiment of the technology includes a substrate

including a reference plane; a support member disposed on the substrate, the support member including at least one inclined surface inclined with respect to the reference plane; at least one magnetic detection element disposed on the at least one inclined surface; a first insulating portion of an insulating material disposed on a part of the at least one magnetic detection element; and a second insulating portion of an insulating material disposed on another part of the at least one magnetic detection element at a position forward of the first insulating portion in a direction along the at least one inclined surface, the direction being a direction away from the reference plane.

(7) In the magnetic sensor according to one embodiment of the technology, the first insulating portion is disposed on a part of the magnetic detection element disposed on the inclined surface, and the second insulating portion is disposed on another part of the magnetic detection element. Thereby according to one embodiment of the technology, it is possible to achieve desired characteristics for a magnetic sensor including magnetoresistive elements each disposed on an inclined surface.

(8) Other and further objects, features, and advantages of the technology will appear more fully from the following description.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate example embodiments and, together with the specification, serve to explain the principles of the technology.

(2) FIG. 1 is a perspective view showing a magnetic sensor according to a first example embodiment of the technology.

(3) FIG. 2 is a functional block diagram showing a configuration of a magnetic sensor device including the magnetic sensor according to the first example embodiment of the technology.

(4) FIG. 3 is a circuit diagram showing a circuit configuration of a first detection circuit of the first example embodiment of the technology.

(5) FIG. 4 is a circuit diagram showing a circuit configuration of a second detection circuit of the first example embodiment of the technology.

(6) FIG. 5 is a plan view showing a part of the magnetic sensor according to the first example embodiment of the technology.

(7) FIG. 6 is a sectional view showing a part of the magnetic sensor according to the first example embodiment of the technology.

(8) FIG. 7 is a side view showing a magnetoresistive element of the first example embodiment of the technology.

(9) FIG. 8 is a sectional view showing first and second insulating portions of a first example of the first example embodiment of the technology.

(10) FIG. 9 is a sectional view showing first and second insulating portions of a second example of the first example embodiment of the technology.

(11) FIG. 10 is a sectional view showing first and second insulating portions of a third example of the first example embodiment of the technology.

(12) FIG. 11 is a sectional view showing first and second insulating portions of a fourth example of the first example embodiment of the technology.

(13) FIG. 12 is a sectional view showing a part of the magnetic sensor according to the second example embodiment of the technology.

### DETAILED DESCRIPTION

(14) An object of the technology is to provide a magnetic sensor that includes magnetoresistive elements each disposed on an inclined surface and in which desired characteristics can be achieved.

(15) In the following, some example embodiments and modification examples of the technology are described in detail with reference to the accompanying drawings. Note that the following description is directed to illustrative examples of the disclosure and not to be construed as limiting the technology. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting the technology. Further, elements in the following example embodiments which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Like elements are denoted with the same reference numerals to avoid redundant descriptions. Note that the description is given in the following order.

#### First Example Embodiment

(16) First, a configuration of a magnetic sensor according to a first example embodiment of the technology will be described with reference to FIGS. 1 and 2. FIG. 1 is a perspective view showing a magnetic sensor according to the example embodiment. FIG. 2 is a functional block diagram showing a configuration of a magnetic sensor device including the magnetic sensor according to the example embodiment.

(17) As shown in FIG. 1, the magnetic sensor 1 is in the form of a chip having a rectangular parallelepiped shape. The magnetic sensor 1 includes a top surface 1a and a bottom surface located opposite to each other and also includes four side surfaces connecting the top surface 1a to the bottom surface. The magnetic sensor 1 also includes a plurality of electrode pads disposed on the top surface 1a.

(18) Now, a description will be given of a reference coordinate system in the present example embodiment with reference to FIG. 1. The reference coordinate system is an orthogonal coordinate system that is set with reference to a magnetic sensor 1 and defined by three axes. An X direction, a Y direction, and a Z direction are defined in the reference coordinate system. The X, Y, and Z directions are orthogonal to each other. In particular, in the example embodiment, a direction that is perpendicular to the top surface 1a of the magnetic sensor 1 and is oriented from the bottom surface to the top surface 1a of the magnetic sensor 1 is defined as the Z direction. The opposite directions to the X, Y, and Z directions will be expressed as -X, -Y, and -Z directions, respectively. The three axes defining the reference coordinate system are an axis parallel to the X direction, an axis parallel to the Y direction, and an axis parallel to the Z direction.

(19) Hereinafter, the term “above” refers to positions located forward of a reference position in the Z direction, and “below” refers to positions opposite from the “above” positions with respect to the reference position. For each component of the magnetic sensor 1, the term “top surface” refers to a surface of the component located at the end thereof in the Z direction, and “bottom surface” refers to a surface of the component located at the end thereof in the -Z direction. The phrase “when seen in the Z direction” means that an object is seen from a position at a distance in the Z direction.

(20) As shown in FIG. 2, the magnetic sensor 1 includes a first detection circuit 20 and a second detection circuit 30. Each of the first and second detection circuits 20 and 30 includes a plurality of magnetic detection elements, and is configured to detect a target magnetic field to generate at least one detection signal. In particular, in the example embodiment, the plurality of magnetic detection elements are a plurality of magnetoresistive elements. The magnetoresistive elements will hereinafter be referred to as MR elements.

(21) A plurality of detection signals generated by the first and second detection circuits 20 and 30 are processed by a processor 40. The magnetic sensor 1 and the processor 40 constitute a magnetic sensor device 100. The processor 40 is configured to, by processing the plurality of detection signals generated by the first and second detection circuits 20 and 30, generate a first detection value and a second detection value respectively having correspondences with components of a magnetic field in two different directions at a predetermined reference position. In particular, in the present example embodiment, the foregoing two different directions are a direction parallel to an

XY plane and a direction parallel to the Z direction. For example, the processor **40** is constructed of an application-specific integrated circuit (ASIC).

(22) The processor **40** may be included in a support supporting the magnetic sensor **1**, for example. The support includes a plurality of electrode pads. The first and second detection circuits **20** and **30** are connected to the processor **40** via the plurality of electrode pads of the magnetic sensor **1**, the plurality of electrode pads of the support, and a plurality of bonding wires, for example. In a case where the plurality of electrode pads of the magnetic sensor **1** are provided on the top surface **1a** of the magnetic sensor **1**, the magnetic sensor **1** may be mounted on the top surface of the support in such a posture that the bottom surface of the magnetic sensor **1** faces the top surface of the support.

(23) Next, the first and second detection circuits **20** and **30** will be described with reference to FIGS. **3** to **6**. FIG. **3** is a circuit diagram showing a circuit configuration of the first detection circuit **20**. FIG. **4** is a circuit diagram showing a circuit configuration of the second detection circuit **30**. FIG. **5** is a plan view showing a part of the magnetic sensor **1**. FIG. **6** is a sectional view showing a part of the magnetic sensor **1**.

(24) Here, as shown in FIG. **5**, a U direction and a V direction are defined as follows. The U direction is a direction rotated from the X direction to the  $-Y$  direction. The V direction is a direction rotated from the Y direction to the X direction. More specifically, in the present example embodiment, the U direction is set to a direction rotated from the X direction to the  $-Y$  direction by  $\alpha$ , and the V direction is set to a direction rotated from the Y direction to the X direction by  $\alpha$ . Note that  $\alpha$  is an angle greater than  $0^\circ$  and smaller than  $90^\circ$ . For example,  $\alpha$  is  $45^\circ$ .  $-U$  direction refers to a direction opposite to the U direction, and  $-V$  direction refers to a direction opposite to the V direction.

(25) As shown in FIG. **6**, a W1 direction and a W2 direction are defined as follows. The W1 direction is a direction rotated from the V direction to the  $-Z$  direction. The W2 direction is a direction rotated from the V direction to the Z direction. More specifically, in the present example embodiment, the W1 direction is set to a direction rotated from the V direction to the  $-Z$  direction by  $\beta$ , and the W2 direction is set to a direction rotated from the V direction to the Z direction by  $\beta$ . Note that  $\beta$  is an angle greater than  $0^\circ$  and smaller than  $90^\circ$ .  $-W1$  direction refers to a direction opposite to the W1 direction, and  $-W2$  direction refers to a direction opposite to the W2 direction. The W1 direction and W2 direction both are orthogonal to the U direction.

(26) The first detection circuit **20** is configured to detect a component of the target magnetic field in a direction parallel to the W1 direction and generate at least one first detection signal which has a correspondence with the component. The second detection circuit **30** is configured to detect a component of the target magnetic field in a direction parallel to the W2 direction and generate at least one second detection signal which has a correspondence with the component.

(27) As shown in FIG. **3**, the first detection circuit **20** includes a power supply port V2, a ground port G2, signal output ports E21 and E22, a first resistor section R21, a second resistor section R22, a third resistor section R23, and a fourth resistor section R24. The plurality of MR elements of the first detection circuit **20** constitute the first to fourth resistor sections R21, R22, R23, and R24.

(28) The first resistor section R21 is provided between the power supply port V2 and the signal output port E21. The second resistor section R22 is provided between the signal output port E21 and the ground port G2. The third resistor section R23 is provided between the signal output port E22 and the ground port G2. The fourth resistor section R24 is provided between the power supply port V2 and the signal output port E22.

(29) As shown in FIG. **4**, the second detection circuit **30** includes a power supply port V3, a ground port G3, signal output ports E31 and E32, a first resistor section R31, a second resistor section R32, a third resistor section R33, and a fourth resistor section R34. The plurality of MR elements of the second detection circuit **30** constitute the first to fourth resistor sections R31, R32, R33, and R34.

(30) The first resistor section R31 is provided between the power supply port V3 and the signal output port E31. The second resistor section R32 is provided between the signal output port E31

and the ground port G3. The third resistor section R33 is provided between the signal output port E32 and the ground port G3. The fourth resistor section R34 is provided between the power supply port V3 and the signal output port E32.

(31) A voltage or current of a predetermined magnitude is applied to each of the power supply ports V2 and V3. Each of the ground ports G2 and G3 is connected to the ground.

(32) The plurality of MR elements of the first detection circuit 20 will be referred to as a plurality of first MR elements 50B. The plurality of MR elements of the second detection circuit 30 will be referred to as a plurality of second MR elements 50C. Since the first and second detection circuits 20 and 30 are the components of the magnetic sensor 1, it can be said that the magnetic sensor 1 includes the plurality of first MR elements 50B and the plurality of second MR elements 50C. Any given MR element will be denoted by the reference numeral 50.

(33) FIG. 7 is a side view showing the MR element 50. The MR element 50 is a spin-valve MR element including a plurality of magnetic layers. The MR element 50 includes a magnetization pinned layer 51 having a magnetization whose direction is fixed, a free layer 53 having a magnetization whose direction is variable depending on the direction of a target magnetic field, and a gap layer 52 located between the magnetization pinned layer 51 and the free layer 53. The MR element 50 may be a tunneling magnetoresistive (TMR) element or a giant magnetoresistive (GMR) element. In the TMR element, the gap layer 52 is a tunnel barrier layer. In the GMR element, the gap layer 52 is a nonmagnetic conductive layer. The resistance of the MR element 50 changes with the angle that the magnetization direction of the free layer 53 forms with respect to the magnetization direction of the magnetization pinned layer 51. The resistance of the MR element 50 is at its minimum value when the foregoing angle is 0°, and at its maximum value when the foregoing angle is 180°. In each MR element 50, the free layer 53 has a shape anisotropy that sets the direction of the magnetization easy axis to be orthogonal to the magnetization direction of the magnetization pinned layer 51. As a method for setting the magnetization easy axis in a predetermined direction in the free layer 53, a magnet configured to apply a bias magnetic field to the free layer 53 can be used. The magnetization pinned layer 51, the gap layer 52, and the free layer 53 are stacked in this order.

(34) The MR element 50 may further include an antiferromagnetic layer disposed on the magnetization pinned layer 51 on the side opposite to the gap layer 52. The antiferromagnetic layer is formed of an antiferromagnetic material, and is in exchange coupling with the magnetization pinned layer 51 to thereby pin the magnetization direction of the magnetization pinned layer 51. Alternatively, the magnetization pinned layer 51 may be a so-called self-pinned layer (Synthetic Ferri Pinned layer, SFP layer). The self-pinned layer has a stacked ferri structure in which a ferromagnetic layer, a nonmagnetic intermediate layer, and a ferromagnetic layer are stacked, and the two ferromagnetic layers are antiferromagnetically coupled.

(35) It should be appreciated that the layers 51 to 53 of each MR element 50 may be stacked in the reverse order to that shown in FIG. 7.

(36) In FIGS. 3 and 4, solid arrows represent the magnetization directions of the magnetization pinned layers 51 of the MR elements 50. Hollow arrows represent the magnetization directions of the free layers 53 of the MR elements 50 in a case where no target magnetic field is applied to the MR elements 50.

(37) In the example shown in FIG. 3, the magnetization directions of the magnetization pinned layers 51 in each of the first and third resistor sections R21 and R23 are the W1 direction. The magnetization directions of the magnetization pinned layers 51 in each of the second and fourth resistor sections R22 and R24 are the -W1 direction. The free layer 53 in each of the plurality of first MR elements 50B has a shape anisotropy that sets the direction of the magnetization easy axis to a direction parallel to the U direction. The magnetization directions of the free layers 53 in each of the first and second resistor sections R21 and R22 in a case where no target magnetic field is applied to the first MR elements 50B are the U direction. The magnetization directions of the free

layers **53** in each of the third and fourth resistor sections **R23** and **R24** in the foregoing case are the  $-U$  direction.

(38) In the example shown in FIG. 4, the magnetization directions of the magnetization pinned layers **51** in each of the first and third resistor sections **R31** and **R33** are the  $W2$  direction. The magnetization directions of the magnetization pinned layers **51** in each of the second and fourth resistor sections **R32** and **R34** are the  $-W2$  direction. The free layer **53** in each of the plurality of second MR elements **50C** has a shape anisotropy that sets the direction of the magnetization easy axis to a direction parallel to the  $U$  direction. The magnetization directions of the free layers **53** in each of the first and second resistor sections **R31** and **R32** in a case where no target magnetic field is applied to the second MR elements **50C** are the  $U$  direction. The magnetization directions of the free layers **53** in each of the third and fourth resistor sections **R33** and **R34** in the foregoing case are the  $-U$  direction.

(39) The magnetic sensor **1** includes a magnetic field generator configured to apply a magnetic field in a predetermined direction to the free layer **53** of each of the plurality of first MR elements **50B**, and the plurality of second MR elements **50C**. In the present example embodiment, the magnetic field generator includes a coil **80** that applies a magnetic field in the predetermined direction to the free layer **53** in each of the plurality of first MR elements **50B** and the plurality of second MR elements **50C**.

(40) Note that the magnetization directions of the magnetization pinned layers **51** and the directions of the magnetization easy axes of the free layers **53** may slightly deviate from the foregoing directions from the perspective of the accuracy of the manufacturing of the MR elements **50** and the like. The magnetization pinned layers **51** may be magnetized to include magnetization components in the foregoing directions as their main components. In such a case, the magnetization directions of the magnetization pinned layers **51** are the same or substantially the same as the foregoing directions.

(41) In the present example embodiment, the MR element **50** is configured such that a current flows in the stacking direction of the plurality of magnetic layers, that is, the magnetization pinned layer **51** and the free layer **53**. As described below, the magnetic sensor **1** includes a lower electrode and an upper electrode for flowing a current through the MR element **50**. The MR element **50** is disposed between the lower electrode and the upper electrode.

(42) Hereinafter, a specific structure of the magnetic sensor **1** will be described in detail with reference to FIGS. 5 and 6. FIG. 6 shows a part of a cross section at a position indicated by the line 6-6 in FIG. 5.

(43) The magnetic sensor **1** includes a substrate **301** with a top surface **301a**, insulating layers **302**, **303**, **304**, **305**, **306**, **307**, **308**, **309**, and **310**, a plurality of lower electrodes **61B**, a plurality of lower electrodes **61C**, a plurality of upper electrodes **62B**, a plurality of upper electrodes **62C**, a plurality of lower coil elements **81**, and a plurality of upper coil elements **82**. It is assumed that the top surface **301a** of the substrate **301** is parallel to the  $XY$  plane. The  $Z$  direction is also a direction perpendicular to the top surface **301a** of the substrate **301**. The coil elements are a part of the coil winding.

(44) The insulating layer **302** is disposed on the substrate **301**. The plurality of lower coil elements **81** are disposed on the insulating layer **302**. The insulating layer **303** is disposed around the plurality of lower coil elements **81** on the insulating layer **302**. The insulating layers **304** and **305** are stacked in this order on the plurality of lower coil elements **81** and the insulating layer **303**.

(45) The plurality of lower electrodes **61B** and the plurality of lower electrodes **61C** are disposed on the insulating layer **305**. The plurality of first MR elements **50B** are disposed on the plurality of lower electrodes **61B**. The plurality of second MR elements **50C** are disposed on the plurality of lower electrodes **61C**. The insulating layer **306** is disposed on the plurality of lower electrodes **61B** and the plurality of lower electrodes **61C** and around the plurality of first MR elements **50B** and around the plurality of second MR elements **50C**. The insulating layer **307** is disposed on the



insulating layer **305** and around the plurality of lower electrodes **61B**, around the plurality of lower electrodes **61C**, and around the insulating layer **306**.

(46) The insulating layer **308** is disposed on a part of each of the plurality of first MR elements **50B**, on a part of each of the plurality of second MR elements **50C**, and on the insulating layers **306** and **307**. The plurality of upper electrodes **62B** are disposed on another part of each of the plurality of first MR elements **50B** and on a part of the insulating layer **308**. The plurality of upper electrodes **62C** are disposed on another part of each of the plurality of second MR elements **50C** and on a part of the insulating layer **308**. The insulating layer **309** is disposed on another part of the insulating layer **308** and around the plurality of upper electrodes **62B** and around the plurality of upper electrodes **62C**.

(47) The insulating layer **310** is disposed on the plurality of upper electrodes **62B**, the plurality of upper electrodes **62C**, and the insulating layer **309**. The plurality of upper coil elements **82** are disposed on the insulating layer **310**. The magnetic sensor **1** may further include a not-shown insulating layer that covers the plurality of upper coil elements **82** and the insulating layer **310**.

(48) The magnetic sensor **1** includes a support member supporting the plurality of first MR elements **50B** and the plurality of second MR elements **50C**. The support member includes at least one inclined surface inclined with respect to the top surface **301a** of the substrate **301**. In particular, in the example embodiment, the support member includes the insulating layer **305**. Note that FIG. **5** shows the insulating layer **305**, the plurality of first MR elements **50B**, the plurality of second MR elements **50C**, and the plurality of upper coil elements **82** among the components of the magnetic sensor **1**.

(49) The insulating layer **305** includes a plurality of protruding surfaces **305c** each protruding in a direction (the Z direction) away from the top surface **301a** of the substrate **301**. Each of the plurality of protruding surfaces **305c** extends in a direction parallel to the U direction. The overall shape of each of the protruding surfaces **305c** is a semi-cylindrical curved surface formed by moving the curved shape (arch shape) of the protruding surface **305c** shown in FIG. **6** along the direction parallel to the U direction. The plurality of protruding surfaces **305c** are arranged at predetermined intervals along a direction parallel to the V direction.

(50) Each of the plurality of protruding surfaces **305c** includes an upper end portion farthest from the top surface **301a** of the substrate **301**. In the example embodiment, each of the upper end portions of the plurality of protruding surfaces **305c** extends in the direction parallel to the U direction. Herein, focus is placed on a given protruding surface **305c** of the plurality of protruding surfaces **305c**. The protruding surface **305c** includes a first inclined surface **305a** and a second inclined surface **305b**. The first inclined surface **305a** refers to the part of the protruding surface **305c** on the side of the V direction of the upper end portion of the protruding surface **305c**. The second inclined surface **305b** refers to the part of the protruding surface **305c** on the side of the -V direction of the upper end portion of the protruding surface **305c**. In FIG. **5**, a boundary between the first inclined surface **305a** and the second inclined surface **305b** is indicated by a dotted line.

(51) The upper end portion of the protruding surface **305c** may be the boundary between the first inclined surface **305a** and the second inclined surface **305b**. In such a case, the dotted line shown in FIG. **5** indicates the upper end portion of the protruding surface **305c**.

(52) The top surface **301a** of the substrate **301** is parallel to the XY plane. Each of the first inclined surface **305a** and the second inclined surface **305b** is inclined with respect to the top surface **301a** of the substrate **301**, that is, the XY plane. In a cross section perpendicular to the top surface **301a** of the substrate **301**, a distance between the first inclined surface **305a** and the second inclined surface **305b** becomes smaller in a direction away from the top surface **301a** of the substrate **301**.

(53) In the example embodiment, since two or more protruding surface **305c** are present, the number of each of the first inclined surfaces **305a** and the second inclined surfaces **305b** is also two or more. The insulating layer **305** includes the plurality of first inclined surfaces **305a** and the plurality of second inclined surfaces **305b**.

(54) The insulating layer **305** further includes a flat surface **305d** present around the plurality of protruding surfaces **305c**. The flat surface **305d** is a surface parallel to the top surface **301a** of the substrate **301**. Each of the plurality of protruding surfaces **305c** protrudes in the Z direction from the flat surface **305d**. In the example embodiment, the plurality of protruding surfaces **305c** are disposed at predetermined intervals. Thus, the flat surface **305d** is present between the two protruding surfaces **305c** adjoining in the V direction.

(55) The insulating layer **305** includes a plurality of protruding portions each protruding in the Z direction, and a flat portion present around the plurality of protruding portions. Each of the plurality of protruding portions extends in the direction parallel to the U direction and includes the protruding surface **305c**. The plurality of protruding portions are arranged at predetermined intervals in the direction parallel to the V direction. The thickness (the dimension in the Z direction) of the flat portion is substantially constant. The insulating layer **304** has a substantially constant thickness (i.e., a dimension in the Z direction), and is formed along the bottom surface of the insulating layer **305**.

(56) The plurality of lower electrodes **61B** are disposed on the plurality of first inclined surfaces **305a**. The plurality of lower electrodes **61C** are disposed on the plurality of second inclined surfaces **305b**. As described above, since each of the first inclined surfaces **305a** and the second inclined surfaces **305b** is inclined with respect to the top surface **301a** of the substrate **301**, that is, the XY plane, each of the top surfaces of the plurality of lower electrodes **61B** and each of the top surfaces of the plurality of lower electrodes **61C** are also inclined with respect to the XY plane. Thus, it can be said that the plurality of first MR elements **50B** and the plurality of second MR elements **50C** are disposed on the inclined surfaces inclined with respect to the XY plane. The insulating layer **305** is a member for supporting each of the plurality of first MR elements **50B** and the plurality of second MR elements **50C** so as to allow such MR elements to be inclined with respect to the XY plane.

(57) Note that in the example embodiment, the first inclined surfaces **305a** are curved surfaces. Therefore, the first MR elements **50B** are curved along the curved surfaces (the first inclined surfaces **305a**). For the sake of convenience, in the present example embodiment, the magnetization directions of the magnetization pinned layers **51** of the first MR elements **50B** are defined as straight directions as described above. The W1 direction and the -W1 direction that are the magnetization directions of the magnetization pinned layers **51** of the first MR elements **50B** are also directions in which the tangents to the first inclined surfaces **305a** at the vicinity of the first MR elements **50B** extend.

(58) Similarly, in the example embodiment, the second inclined surfaces **305b** are curved surfaces. Therefore, the second MR elements **50C** are curved along the curved surfaces (the second inclined surfaces **305b**). For the sake of convenience, in the present example embodiment, the magnetization directions of the magnetization pinned layers **51** of the second MR elements **50C** are defined as straight directions as described above. The W2 direction and the -W2 direction that are the magnetization directions of the magnetization pinned layers **51** of the second MR elements **50C** are also directions in which the tangents to the second inclined surfaces **305b** at the vicinity of the second MR elements **50C** extend.

(59) As shown in FIG. 5, the plurality of first MR elements **50B** are disposed so that two or more MR elements **50B** are arranged both in the U direction and in the V direction. The plurality of first MR elements **50B** are aligned in a row on one first inclined surface **305a**. Similarly, the plurality of second MR elements **50C** are disposed so that two or more MR elements **50C** are arranged both in the U direction and in the V direction. The plurality of second MR elements **50C** are aligned in a row on one second inclined surface **305b**. In the example embodiment, the row of the plurality of first MR elements **50B** and the row of the plurality of second MR elements **50C** are alternately arranged in the direction parallel to the V direction.

(60) Note that one first MR element **50B** and one second MR element **50C** adjoining each other

may or may not deviate in the direction parallel to the U direction when seen in the Z direction. Two first MR elements **50B** adjoining each other across one second MR element **50C** may or may not deviate in the direction parallel to the U direction when seen in the Z direction. Two second MR elements **50C** adjoining each other across one first MR element **50B** may or may not deviate in the direction parallel to the U direction when seen in the Z direction.

(61) The plurality of first MR elements **50B** are connected in series by the plurality of lower electrodes **61B** and the plurality of upper electrodes **62B**. Herein, a method for connecting the plurality of first MR elements **50B** will be described in detail with reference to FIG. 7. In FIG. 7, the reference sign **61** denotes a lower electrode corresponding to a given MR element **50**, and the reference numeral **62** denotes an upper electrode corresponding to the given MR element **50**. As shown in FIG. 7, each lower electrode **61** has a long slender shape. Two lower electrodes **61** adjoining in the longitudinal direction of the lower electrodes **61** have a gap therebetween. MR elements **50** are disposed near both longitudinal ends on the top surface of each lower electrode **61**. Each upper electrode **62** has a long slender shape, and electrically connects two adjoining MR elements **50** that are disposed on two lower electrodes **61** adjoining in the longitudinal direction of the lower electrodes **61**.

(62) Although not shown, one MR element **50** located at the end of a row of a plurality of aligned MR elements **50** is connected to another MR element **50** located at the end of another row of a plurality of MR elements **50** adjoining in a direction intersecting with the longitudinal direction of the lower electrodes **61**. Such two MR elements **50** are connected to each other by a not-shown electrode. The not-shown electrode may be an electrode that connects the bottom surfaces or the top surfaces of the two MR elements **50**.

(63) In a case where the MR elements **50** shown in FIG. 7 are the first MR elements **50B**, the lower electrodes **61** shown in FIG. 7 correspond to the lower electrodes **61B**, and the upper electrodes **62** shown in FIG. 7 correspond to the upper electrodes **62B**. In such a case, the longitudinal direction of the lower electrodes **61** is parallel to the U direction.

(64) Similarly, the plurality of second MR elements **50C** are connected in series by the plurality of lower electrodes **61C** and the plurality of upper electrodes **62C**. The foregoing description of the method for connecting the plurality of first MR elements **50B** holds true also for the method for connecting the plurality of second MR elements **50C**. In a case where the MR elements **50** shown in FIG. 7 are the second MR elements **50C**, the lower electrodes **61** shown in FIG. 7 correspond to the lower electrodes **61C**, and the upper electrodes **62** shown in FIG. 7 correspond to the upper electrodes **62C**. In such a case, the longitudinal direction of the lower electrodes **61** is parallel to the U direction.

(65) Each of the plurality of upper coil elements **82** extends in a direction parallel to the Y direction. The plurality of upper coil elements **82** are arranged in the X direction. In particular, in the present example embodiment, when seen in the Z direction, each of the plurality of first MR elements **50B** and the plurality of second MR elements **50C** overlaps two upper coil elements **82**.

(66) Each of the plurality of lower coil elements **81** extends in a direction parallel to the Y direction. The plurality of lower coil elements **81** are arranged in the X direction. The shape and arrangement of the plurality of lower coil elements **81** may be the same as or different from those of the plurality of upper coil elements **82**. In the example shown in FIGS. 5 and 6, the dimension in the X direction of each of the plurality of lower coil elements **81** is smaller than the dimension in the X direction of each of the plurality of upper coil elements **82**. The distance between two lower coil elements **81** adjoining in the X direction is smaller than the distance between two upper coil elements **82** adjoining in the X direction.

(67) In the example shown in FIGS. 5 and 6, the plurality of lower coil elements **81** and the plurality of upper coil elements **82** are electrically connected so as to constitute the coil **80** that applies a magnetic field in a direction parallel to the X direction to the free layer **53** in each of the plurality of first MR elements **50B** and the plurality of second MR elements **50C**. Alternatively, the

coil **80** may be configured to be able to, for example, apply a magnetic field in the X direction to the free layers **53** in the first and second resistor sections **R21** and **R22** of the first detection circuit **20** and the first and second resistor sections **R31** and **R32** of the second detection circuit **30**, and apply a magnetic field in the -X direction to the free layers **53** in the third and fourth resistor sections **R23** and **R24** of the first detection circuit **20** and the third and fourth resistor sections **R33** and **R34** of the second detection circuit **30**. The coil **80** may be controlled by the processor **40**.

(68) Next, the first and second detection signals will be described. First, the first detection signal will be described with reference to FIG. 3. As the strength of the component of the target magnetic field in the direction parallel to the W1 direction changes, the resistance of each of the resistor sections **R21** to **R24** of the first detection circuit **20** changes either so that the resistances of the resistor sections **R21** and **R23** increase and the resistances of the resistor sections **R22** and **R24** decrease or so that the resistances of the resistor sections **R21** and **R23** decrease and the resistances of the resistor sections **R22** and **R24** increase. Thereby the electric potential of each of the signal output ports **E21** and **E22** changes. The first detection circuit **20** generates a signal corresponding to the electric potential of the signal output port **E21** as a first detection signal **S21**, and generates a signal corresponding to the electric potential of the signal output port **E22** as a first detection signal **S22**.

(69) Next, the second detection signal will be described with reference to FIG. 4. As the strength of the component of the target magnetic field in the direction parallel to the W2 direction changes, the resistance of each of the resistor sections **R31** to **R34** of the second detection circuit **30** changes either so that the resistances of the resistor sections **R31** and **R33** increase and the resistances of the resistor sections **R32** and **R34** decrease or so that the resistances of the resistor sections **R31** and **R33** decrease and the resistances of the resistor sections **R32** and **R34** increase. Thereby the electric potential of each of the signal output ports **E31** and **E32** changes. The second detection circuit **30** generates a signal corresponding to the electric potential of the signal output port **E31** as a second detection signal **S31**, and generates a signal corresponding to the electric potential of the signal output port **E32** as a second detection signal **S32**.

(70) Next, the operation of the processor **40** will be described. The processor **40** is configured to generate the first detection value and the second detection value based on the first detection signals **S21** and **S22** and the second detection signals **S31** and **S32**. The first detection value is a detection value corresponding to the component of the target magnetic field in the direction parallel to the V direction. The second detection value is a detection value corresponding to the component of the target magnetic field in the direction parallel to the Z direction. The first detection value is represented by a symbol  $S_v$ , and the second detection value is represented by a symbol  $S_z$ .

(71) The processor **40** generates the first and second detection values  $S_v$  and  $S_z$  as follows, for example. First, the processor **40** generates a value **S1** by an arithmetic including obtainment of the difference **S21-S22** between the first detection signal **S21** and the first detection signal **S22**, and generates a value **S2** by an arithmetic including obtainment of the difference **S31-S32** between the second detection signal **S31** and the second detection signal **S32**. Next, the processor **40** calculates values **S3** and **S4** using the following expressions (1) and (2).

$$S3=(S2+S1)/(2 \cos \alpha) \quad (1)$$

$$S4=(S2-S1)/(2 \sin \alpha) \quad (2)$$

(72) The first detection value  $S_v$  may be the value **S3** itself, or may be a result of a predetermined correction, such as a gain adjustment or an offset adjustment, made to the value **S3**. In the same manner, the second detection value  $S_z$  may be the value **S4** itself, or may be a result of a predetermined correction, such as a gain adjustment or an offset adjustment, made to the value **S4**.

(73) Next, features of the structure of the magnetic sensor **1** according to the example embodiment will be described. First, a first example will be described. FIG. 8 is a sectional view showing first and second insulating portions of the first example.

(74) FIG. 8 shows a cross section crossing the MR element **50** disposed on a given inclined surface

**305e** and parallel to a VZ plane. Hereinafter, a cross section parallel to the VZ plane will be referred to as a VZ cross section. The VZ cross section shown in FIG. **8** may be the one in which a cross section of the MR element **50** is seen from a position on the side of the U direction as in FIG. **6**. In such a case, the MR element **50**, the lower electrode **61**, and the inclined surface **305e** respectively correspond to the first MR element **50B**, the lower electrode **61B**, and the first inclined surface **305a**. Alternatively, the VZ cross section shown in FIG. **8** may be the one in which a cross section of the MR element **50** is seen from a position on the side of the -U direction. In such a case, the MR element **50**, the lower electrode **61**, and the inclined surface **305e** respectively correspond to the second MR element **50C**, the lower electrode **61C**, and the second inclined surface **305b**.

(75) Herein, as shown in FIGS. **8** and **9**, a first direction **D1** and a second direction **D2** parallel to the VZ plane are defined. The first direction **D1** is a direction lying along the inclined surface **305e** and is also a direction away from a reference plane. In the present example embodiment, it is assumed that the top surface **301a** of the substrate **301** (see FIG. **6**) is the reference plane. The Z direction is one direction perpendicular to the reference plane (the top surface **301a** of the substrate **301**). The second direction **D2** is a direction lying along the inclined surface **305e** and is also a direction closer to the reference plane (the top surface **301a** of the substrate **301**).

(76) In the following description, a direction that lies along the inclined surface **305e** and is parallel to the first direction **D1** (a direction parallel to the second direction **D2**) will be simply referred to as a direction along the inclined surface **305e**. Such a direction is also a direction that lies along the inclined surface **305e** and in which a distance from the reference plane (the top surface **301a** of the substrate **301**) changes.

(77) The MR element **50** includes a bottom surface **50a** facing the inclined surface **305e**, a top surface **50b** on the side opposite to the bottom surface **50a**, a first side surface **50c**, and a second side surface **50d**. The first side surface **50c** connects an end portion of the bottom surface **50a** on the side of the second direction **D2** and an end portion of the top surface **50b** on the side of the second direction **D2**. The second side surface **50d** is disposed forward of the first side surface **50c** in the first direction **D1**. The second side surface **50d** connects an end portion of the bottom surface **50a** on the side of the first direction **D1** and an end portion of the top surface **50b** on the side of the first direction **D1**.

(78) The lower electrode **61** is provided between the MR element **50** and the inclined surface **305e**. The lower electrode **61** includes a bottom surface **61a** facing the inclined surface **305e**, a top surface **61b** on a side opposite to the bottom surface **61a**, and two side surfaces (see FIG. **6**) each connecting the bottom surface **61a** and the top surface **61b**. Note that the lower electrode **61** may be formed on a region from the inclined surface **305e** to the flat surface **305d**. In such a case, one of the two side surfaces of the lower electrode **61** is disposed on the inclined surface **305e** and the other is disposed on the flat surface **305d**. Alternatively, the entire lower electrode **61** may be disposed on the inclined surface **305e**. In such a case, both the two side surfaces of the lower electrode **61** are disposed on the inclined surface **305e**.

(79) The magnetic sensor **1** includes a first insulating portion **311** and a second insulating portion **312**. Each of the first and second insulating portions **311** and **312** may include a single insulating layer or a plurality of insulating layers. In particular, in the present example embodiment, each of the first and second insulating portions **311** and **312** includes the insulating layers **306** and **308** shown in FIG. **6**. Note that each of the insulating layers **306** and **308** may also include a single insulating film or a plurality of insulating films.

(80) Since each of the insulating layers **306** and **308** is formed of an insulating material, each of the first and second insulating portions **311** and **312** is also formed of an insulating material.

Al.sub.2O.sub.3 or SiO.sub.2, for example, is used as an insulating material for forming each of the first and second insulating portions **311** and **312** (each of the insulating layers **306** and **308**).

(81) The first insulating portion **311** is disposed on a part of the MR element **50**. In particular, in the

present example embodiment, the first insulating portion **311** is disposed on the first side surface **50c** of the MR element **50** and on a part of the top surface **50b** of the MR element **50**.

(82) The second insulating portion **312** is disposed on another part of the MR element **50** at a position forward of the first insulating portion **311** in the first direction **D1**. In particular, in the present example embodiment, the second insulating portion **312** is disposed on the second side surface **50d** of the MR element **50** and on another part of the top surface **50b** of the MR element **50**.

(83) The upper electrode **62** (see FIG. 7) is disposed on the MR element **50**, the first insulating portion **311**, and the second insulating portion **312**, and is electrically connected to the MR element **50**. A part of each of the first and second insulating portions **311** and **312** is provided between the MR element **50** and the upper electrode **62**. Another part of each of the first and second insulating portions **311** and **312** is provided between the lower electrode **61** and the upper electrode **62**.

(84) The description has been made heretofore of the structural features of the magnetic sensor **1** focusing on a single inclined surface **305e** (a single first inclined surface **305a** or a single second inclined surface **305b**). In the present example embodiment, there are a plurality of first inclined surfaces **305a** and a plurality of second inclined surfaces **305b**. The foregoing description of the single inclined surface **305e** holds true also for each of the plurality of first inclined surfaces **305a** and the plurality of second inclined surfaces **305b**.

(85) Now, focus attention on a single first inclined surface **305a** and a single second inclined surface **305b** included in a single projecting surface **305c**, a single first MR element **50B** disposed on the single first inclined surface **305a**, and a single second MR element **50C** disposed on the single second inclined surface **305b** (see FIG. 6). The second insulating portion **312** disposed on a part of the first MR element **50B** and the second insulating portion **312** disposed on a part of the second MR element **50C** may be a single continuous undivided insulating portion. In particular, in the present example embodiment, a part, which is continuous, of the insulating layer **308** is formed above a region from the first inclined surface **305a** to the second inclined surface **305b**.

(86) Next, focus attention on the two projecting surfaces **305c** adjoining in the direction parallel to the **V** direction, the first inclined surface **305a** included in the projecting surface **305c** on the side of the  $-V$  direction, the second inclined surface **305b** included in the projecting surface **305c** on the side of the **V** direction, the single first MR element **50B** disposed on the single first inclined surface **305a**, and the single second MR element **50C** disposed on the single second inclined surface **305b** (see FIG. 6). The first insulating portion **311** disposed on a part of the first MR element **50B** and the first insulating portion **311** disposed on a part of the second MR element **50C** may be a single continuous undivided insulating portion. In particular, in the present example embodiment, a part, which is continuous, of the insulating layer **308** is formed above a region from the first inclined surface **305a** to the second inclined surface **305b**.

(87) Though not shown, the first insulating portion **311** and the second insulating portion **312** may also be a single continuous undivided insulating portion. In particular, in the present example embodiment, the insulating layer **308** may be formed above the plurality of first inclined surfaces **305a** and the plurality of second inclined surfaces **305b** without being divided. The insulating layer **306** may be formed around the first MR element **50B** without being divided above each of the plurality of first inclined surfaces **305a**. Similarly, the insulating layer **306** may be formed around the second MR element **50C** without being divided above each of the plurality of second inclined surfaces **305b**.

(88) Next, a second example will be described. FIG. 9 is a sectional view showing the first and second insulating portions **311** and **312** of the second example. In the second example, the area of the top surface **50b** of the MR element **50** covered with the first insulating portion **311** is larger than the area of the top surface **50b** of the MR element **50** covered with the second insulating portion **312**.

(89) Next, a third example will be described. FIG. 10 is a sectional view showing the first and second insulating portions **311** and **312** of the third example. In the third example, the area of the

top surface **50b** of the MR element **50** covered with the second insulating portion **312** is larger than the area of the top surface **50b** of the MR element **50** covered with the first insulating portion **311**. (90) Next, a fourth example will be described. FIG. **11** is a sectional view showing the first and second insulating portions **311** and **312** of the fourth example. In the fourth example, the second insulating portion **312** covers the second side surface **50d** of the MR element **50**, but does not cover the top surface **50b** of the MR element **50**.

(91) Next, operations and effects of the magnetic sensor **1** according to the example embodiment will be described. In the present example embodiment, the insulating layers **306** and **308** are disposed around each MR element **50**. Each of the first and second insulating portions **311** and **312** includes the insulating layers **306** and **308**. The insulating portions (insulating layers) disposed around each MR element **50** are known to influence the characteristics of the MR element **50**. In the present example embodiment, the first insulating portion **311** is disposed on a part of the MR element **50**, and the second insulating portion **312** is disposed on another part of the MR element **50**. Thereby, according to the present example embodiment, desired characteristics can be achieved.

(92) Hereinafter, the sensitivity of the MR element **50** will be described as an example of the characteristics of the MR element **50**. The free layer **53** has a shape anisotropy that sets the direction of the magnetization easy axis to be parallel to the U direction. In a case where a target magnetic field is not applied to the MR element **50**, the magnetization direction of the free layer **53** is the U direction or the -U direction. In the MR element **50** with such a configuration, if the anisotropy of the free layer **53** in a direction orthogonal to the U direction is increased, the magnetization direction of the free layer **53** becomes more likely to change, with the result that the sensitivity of the MR element **50** improves.

(93) For example, it is possible to increase the anisotropy of the free layer **53** in the direction orthogonal to the U direction by forming the free layer **53** using a negative magnetostrictive layer and forming the first and second insulating portions **311** and **312** using an insulating layer that applies a compressive stress to the free layer **53**. In particular, in the present example embodiment, the first insulating portion **311** is disposed on a part of the MR element **50**, and the second insulating portion **312** is disposed on another part of the MR element **50**. In particular, in the present example embodiment, at least one of the first insulating portion **311** or the second insulating portion **312** is disposed on the top surface **50b** of the MR element **50**. Thereby, according to the present example embodiment, the anisotropy of the free layer **53** in the direction orthogonal to the U direction can be increased than a case where the first and second insulating portions **311** and **312** are not disposed on a part of the MR element **50**, and thus, the sensitivity of the MR element **50** can be improved.

(94) By the way, the MR element **50** is formed on the inclined surface **305e**. Each of the first and second side surfaces **50c** and **50d** is tapered due to restrictions on the production process of forming the MR element **50**. Therefore, the area of the free layer **53** located at a position away from the inclined surface **305e** becomes small, and the outer peripheral length of the free layer **53** also becomes short. In a case where the first and second insulating portions **311** and **312** are not disposed on a part of the MR element **50**, it may be impossible to apply a sufficiently high compressive stress to the free layer **53**. In contrast, in the present example embodiment, the first and second insulating portions **311** and **312** are disposed on a part of the MR element **50** as described above. Thereby, according to the present example embodiment, a sufficiently high compressive stress can be applied to the free layer **53**.

(95) Note that the magnitude of the compressive stress applied to the free layer **53** can be controlled by controlling the amounts of the first and second insulating portions **311** and **312** disposed on each MR element **50** or adjusting the configuration of each of the first and second insulating portions **311** and **312**. For example, in a case where each of the first and second insulating portions **311** and **312** has a three-layer structure of

Al.sub.2O.sub.3/SiO.sub.2/Al.sub.2O.sub.3, the magnitude of the compressive stress can be adjusted by changing the proportion of the thickness of each layer.

(96) When the compressive stress applied to the free layer **53** is increased, the sensitivity hysteresis of the MR element **50** may increase. The sensitivity hysteresis of the MR element **50** can be adjusted by controlling the magnitude of the compressive stress applied to the free layer **53** as described above.

(97) Note that the description has been made heretofore of an example case where the free layer **53** is disposed at a position farther from the inclined surface **305e** than a position where the magnetization pinned layer **51** is disposed as shown in FIG. 7. However, the configuration of the MR element **50** is not limited to the example shown in FIG. 7, and the magnetization pinned layer **51** may be disposed at a position farther from the inclined surface **305e** than a position where the free layer **53** is disposed. In such a case, the material of each of the magnetization pinned layer **51**, the first insulating portion **311**, and the second insulating portion **312** may be selected so that the magnetization direction of the magnetization pinned layer **51** becomes less likely to change. In such a case, it is possible to more easily suppress changes in the magnetization direction of the magnetization pinned layer **51** by disposing the first and second insulating portions **311** and **312** on a part of the MR element **50** than a case where the first and second insulating portions **311** and **312** are not disposed on a part of the MR element **50**.

#### Second Example Embodiment

(98) A magnetic sensor according to a second example embodiment of the technology will now be described with reference to FIG. 12. FIG. 12 is a sectional view showing a part of the magnetic sensor **1** according to the present example embodiment.

(99) In the present example embodiment, each of the plurality of protruding surfaces **305c** of the insulating layer **305** has a triangular roof-like overall shape formed by moving the triangular shape of the protruding surface **305c** shown in FIG. 12 in the direction parallel to the U direction. All the plurality of first inclined surfaces **305a** and the plurality of second inclined surfaces **305b** of the insulating layer **305** are flat surfaces. Each of the plurality of first inclined surfaces **305a** is a flat surface parallel to the U direction and the W1 direction. Each of the plurality of second inclined surfaces **305b** is a flat surface parallel to the U direction and the W2 direction.

(100) Like the example shown in FIG. 6, the insulating layer **305** may include a plurality of protrusions for forming the plurality of protruding surfaces **305c**. Alternatively, the insulating layer **305** may include a plurality of slopes arranged in the direction parallel to the V direction. The plurality of slopes each include a first wall surface corresponding to a first inclined surface **305a** and a second wall surface corresponding to a second inclined surface **305b**. A protruding surface **305c** is constituted by the first wall surface of one slope and the second wall surface of another slope adjoining on the -V direction side of the one slope.

(101) In the example shown in FIG. 12, the plurality of slopes each have a bottom surface corresponding to the flat surface **305d**. However, the plurality of slopes do not need to have a bottom surface each.

(102) The configuration, operation, and effects of the present example embodiment are otherwise the same as those of the first example embodiment.

(103) The technology is not limited to the foregoing example embodiments, and various modifications may be made thereto. For example, the shape of each of the first and second insulating portions **311** and **312** is not limited to the example shown in each example embodiment, and may be any shape as long as the requirements of the claims are met.

(104) The magnetic sensor **1** may further include a third detection circuit configured to detect a component of the target magnetic field in a direction parallel to the XY plane, and generate at least one third detection signal having a correspondence with the component. In such a case, the processor **40** may be configured to generate a detection value corresponding to a component of the target magnetic field in the direction parallel to the U direction based on the at least one third



detection signal. The third detection circuit may be integrated with the first and second detection circuits **20** and **30**, or may be included in a chip separate from the first and second detection circuits **20** and **30**.

(105) As described above, the magnetic sensor according to one embodiment of the technology includes a substrate including a reference plane; a support member disposed on the substrate, the support member including at least one inclined surface inclined with respect to the reference plane; at least one magnetic detection element disposed on the at least one inclined surface; a first insulating portion of an insulating material disposed on a part of the at least one magnetic detection element; and a second insulating portion of an insulating material disposed on another part of the at least one magnetic detection element at a position forward of the first insulating portion in a direction along the at least one inclined surface, the direction being a direction away from the reference plane.

(106) The magnetic sensor according to one embodiment of the technology may further include an upper electrode disposed on the at least one magnetic detection element, the first insulating portion, and the second insulating portion, the upper electrode being electrically connected to the at least one magnetic detection element. The magnetic sensor according to one embodiment of the technology may further include a lower electrode provided between the at least one magnetic detection element and the at least one inclined surface, the lower electrode being electrically connected to the at least one magnetic detection element.

(107) In the magnetic sensor according to one embodiment of the technology, the at least one magnetic detection element may include a bottom surface facing the at least one inclined surface, a top surface on a side opposite to the bottom surface, and first and second side surfaces each connecting the bottom surface and the top surface. The first insulating portion may be disposed on at least the first side surface. The second insulating portion may be disposed on at least the second side surface. The first insulating portion may be further disposed on a part of the top surface of the at least one magnetic detection element. The second insulating portion may be further disposed on a part of the top surface of the at least one magnetic detection element. Alternatively, the second insulating portion may not be disposed on the top surface of the at least one magnetic detection element.

(108) In the magnetic sensor according to one embodiment of the technology, the inclined surface may be a curved surface. Alternatively, the at least one inclined surface may be a flat surface.

(109) In the magnetic sensor according to one embodiment of the technology, the at least one inclined surface may include a first inclined surface and a second inclined surface facing different directions. The at least one magnetic detection element may include a first magnetic detection element disposed on the first inclined surface and a second magnetic detection element disposed on the second inclined surface. The support member may include a projecting surface projecting in a direction away from the reference plane. The projecting surface may include the first inclined surface and the second inclined surface. The second insulating portion disposed on the first magnetic detection element and the second insulating portion disposed on the second magnetic detection element may be a single insulating portion.

(110) In the magnetic sensor according to one embodiment of the technology, the at least one inclined surface may include a first inclined surface and a second inclined surface facing different directions. The at least one magnetic detection element may include a first magnetic detection element disposed on the first inclined surface and a second magnetic detection element disposed on the second inclined surface. The support member may include a first projecting surface and a second projecting surface each projecting in a direction away from the reference plane. The first projecting surface may include the first inclined surface. The second projecting surface may include the second inclined surface. The first insulating portion disposed on the first magnetic detection element and the first insulating portion disposed on the second magnetic detection element may be a single insulating portion.

(111) In the magnetic sensor according to one embodiment of the technology, each of the first insulating portion and the second insulating portion may include a first insulating layer of an insulating material and a second insulating layer of an insulating material disposed on the first insulating layer. The at least one magnetic detection element may include a bottom surface facing the at least one inclined surface, a top surface on a side opposite to the bottom surface, and first and second side surfaces each connecting the bottom surface and the top surface. The first insulating layer may be in contact with the first side surface and the second side surface.

(112) Obviously, various modification examples and variations of the technology are possible in the light of the above teachings. Thus, it is to be understood that, within the scope of the appended claims and equivalents thereof, the technology may be practiced in other embodiments than the foregoing example embodiments.

## Claims

1. A magnetic sensor comprising: a substrate including a reference plane; a support member disposed on the substrate, the support member including at least one inclined surface inclined with respect to the reference plane; at least one magnetic detection element disposed on the at least one inclined surface; and an insulating portion of an insulating material disposed on a part of the at least one magnetic detection element.
2. The magnetic sensor according to claim 1, wherein: the insulating portion includes a first portion that is present on a part of the at least one magnetic detection element and a second portion that is not present on the part of the at least one magnetic detection element; the first portion includes a protruding end located at an end of the insulating portion in a direction away from the reference plane; and a thickness of the first portion becomes smaller toward a side opposite to the second portion from the protruding end.
3. The magnetic sensor according to claim 1, wherein: the at least one magnetic detection element includes a bottom surface facing the at least one inclined surface, a top surface on a side opposite to the bottom surface, and first and second side surfaces each connecting the bottom surface and the top surface; the second side surface is disposed forward of the first side surface in a direction along the at least one inclined surface, the direction being a direction away from the reference plane; and the insulating portion is disposed on at least on the first side surface.
4. The magnetic sensor according to claim 3, wherein the insulating portion is further disposed on a part of the top surface of the at least one magnetic detection element.
5. The magnetic sensor according to claim 4, wherein: the insulating portion includes a first portion that is present on a part of the at least one magnetic detection element and a second portion that is not present on the part of the at least one magnetic detection element; the first portion includes a protruding end located at an end of the insulating portion in a direction away from the reference plane; and a thickness of the first portion becomes smaller toward a side opposite to the second portion from the protruding end.
6. The magnetic sensor according to claim 3, wherein the support member includes a protruding surface including the at least one inclined surface.
7. The magnetic sensor according to claim 3, wherein the support member includes a slope including two wall surfaces facing each other, and at least one of the two wall surfaces corresponds to the at least one inclined surface.
8. The magnetic sensor according to claim 1, wherein: the at least one magnetic detection element includes a bottom surface facing the at least one inclined surface, a top surface on a side opposite to the bottom surface, and first and second side surfaces each connecting the bottom surface and the top surface; the second side surface is disposed forward of the first side surface in a direction along the at least one inclined surface, the direction being a direction away from the reference plane; and the insulating portion is disposed on at least on the second side surface.

9. The magnetic sensor according to claim 8, wherein the insulating portion is further disposed on a part of the top surface of the at least one magnetic detection element.
10. The magnetic sensor according to claim 9, wherein: the insulating portion includes a first portion that is present on a part of the at least one magnetic detection element and a second portion that is not present on the part of the at least one magnetic detection element; the first portion includes a protruding end located at an end of the insulating portion in a direction away from the reference plane; and a thickness of the first portion becomes smaller toward a side opposite to the second portion from the protruding end.
11. The magnetic sensor according to claim 8, wherein the insulating portion is not disposed on the top surface of the at least one magnetic detection element.
12. The magnetic sensor according to claim 1, further comprising an upper electrode disposed on the at least one magnetic detection element and the insulating portion, the upper electrode being electrically connected to the at least one magnetic detection element.
13. The magnetic sensor according to claim 1, further comprising a lower electrode provided between the at least one magnetic detection element and the at least one inclined surface, the lower electrode being electrically connected to the at least one magnetic detection element.
14. The magnetic sensor according to claim 1, wherein: the at least one inclined surface includes a first inclined surface and a second inclined surface facing different directions; the at least one magnetic detection element includes a first magnetic detection element disposed on the first inclined surface and a second magnetic detection element disposed on the second inclined surface; the support member includes a projecting surface projecting in a direction away from the reference plane; the projecting surface includes the first inclined surface and the second inclined surface; and the insulating portion disposed on the first magnetic detection element and the insulating portion disposed on the second magnetic detection element are a single insulating portion.
15. The magnetic sensor according to claim 1, wherein: the at least one inclined surface includes a first inclined surface and a second inclined surface facing different directions; the at least one magnetic detection element includes a first magnetic detection element disposed on the first inclined surface and a second magnetic detection element disposed on the second inclined surface; the support member includes a first projecting surface and a second projecting surface each projecting in a direction away from the reference plane; the first projecting surface includes the first inclined surface; the second projecting surface includes the second inclined surface; and the insulating portion disposed on the first magnetic detection element and the insulating portion disposed on the second magnetic detection element are a single insulating portion.
16. The magnetic sensor according to claim 1, wherein the insulating portion includes a first insulating layer of an insulating material and a second insulating layer of an insulating material disposed on the first insulating layer.
-